Sley HORTES

Software Engineer
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EDUCATION

National Dong Hwa University Sep 2023 - Jun 2027
B.S. in Computer Science and Information Engineering
• Relevant coursework: R (Beginner & Intermediate), Big Data, C++, Data Structures, Algorithms, Image Processing, Machine Learning, Scalable Computing, Advanced Computer Vision, Introduction to Generative AI, Distributed Systems, Database Systems.

National Central University Sep 2022 - Aug 2023
Chinese Language in Chinese Language Center
• Studied Mandarin Chinese at the Chinese Language Center, Taoyuan, Taiwan.

EXPERIENCE

Awards

Undergraduate Project Excellent Award May 2026 · Hualien, Taiwan
National Dong Hwa University
• Awarded for Maestro, a two-stream BiLSTM real-time gesture classification system for hands-free presentation control.

Competitions

1st Place, Big Data Competition Dec 2025 · Hualien, Taiwan
National Dong Hwa University
• Classified Moodle engagement patterns using XGBoost. Extracted frequency-domain features via FFT and scalogram analysis. Built a Gemini-powered AI assistant for personalized student and instructor feedback.

Volunteerships

Research Assistant Sep 2026 - Present · Hualien, Taiwan
NDHU CSIE: Intelligent Computing and Mobile Network Lab
• Volunteering as a research assistant.

TECHNICAL SKILLS

Systems & Backend: C++, Go, Python, SQL, Drogon
Full Stack Development: TypeScript, JavaScript, ReactJS, React Native, NodeJS, Qt, Flask, R
Cloud & Data: AWS, GCP, Cloudflare, PostgreSQL, MongoDB, Redis, Docker, Git

LANGUAGES

French (Native) • English (Professional) • Spanish (Intermediate) • Chinese (Intermediate, B1)

PROJECTS

Sigil github.com/HSley13/Sigil
• Local-first C++20 signal daemon with a lock-free signal bus and rule engine connecting sensor/event sources to actions entirely on-device, no cloud round-trip.

HotelOps github.com/HSley13/HotelOps
• Digital operations platform for a hotel chain: real-time room status board, GPS-verified staff clock-in, linen tracking, and cross-device sync across manager and housekeeping apps, replacing Excel sheets and a magnetic scheduling board.

NDHU Task Management github.com/HSley13/ndhu-task-management
• Full-stack mobile task management app with a local-first Expo/React Native client with offline SQLite sync, and a C++ (Drogon) REST API backend with JWT auth and automatic Moodle LMS assignment import.

Maestro github.com/ntsrigaud/Maestro
• Two-stream BiLSTM real-time hand-gesture recognition system for hands-free presentation control, with a MediaPipe Hands feature pipeline, PySide6 control UI, and cross-platform packaged app. Winner of the NDHU Undergraduate Project Excellent Award, 2026.

NDHU Second-Hand Book Store github.com/HSley13/NDHU-SHBS
• Campus marketplace for NDHU students to buy and sell used textbooks. Led the Go/Fiber backend architecture and REST API design end to end, and integrated an AI-based book condition classification model into the pipeline.

JsonFS github.com/HSley13/JsonFS
• C++ FUSE virtual filesystem that mounts JSON files and live REST APIs as a navigable POSIX directory tree, with schema-enforced writes rejected at the syscall layer, atomic write-back, and live URL mounting with TTL refresh.

Robot Arm Load Analyzer github.com/HSley13/robot-arm-analyzer
• Interactive physics + AI tool for robot-arm payload safety: a from-scratch C++ kinematics and dynamics engine (DH kinematics, Jacobian-transpose statics, full Recursive Newton-Euler dynamics), a PyTorch-trained neural surrogate benchmarked against it, and a real-time Qt Quick 3D interface where every joint, link, and payload parameter is a live, draggable control.